Beyond the Needle: Attention-Based Multimodal Deep Learning for Anemia Screening Through Ablation Studies and Comparative Benchmarking

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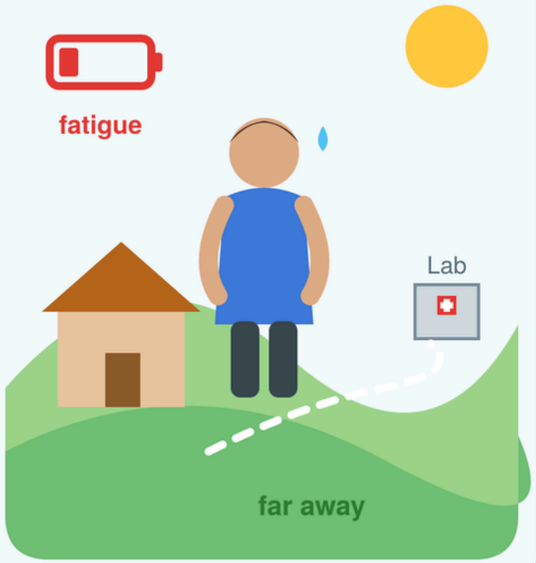
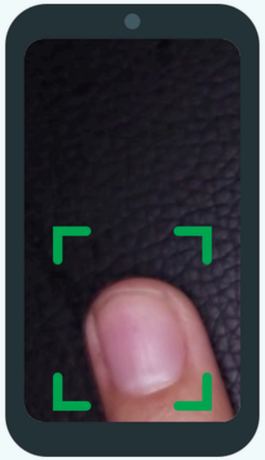
1. Introduction

1.1 Can a Smartphone Camera Help Screen for Anemia

1 · THE PROBLEM



2 · ONE PHOTO



Fingernail colour analysis
by AI, on the phone

3 · QUICK SCREENING

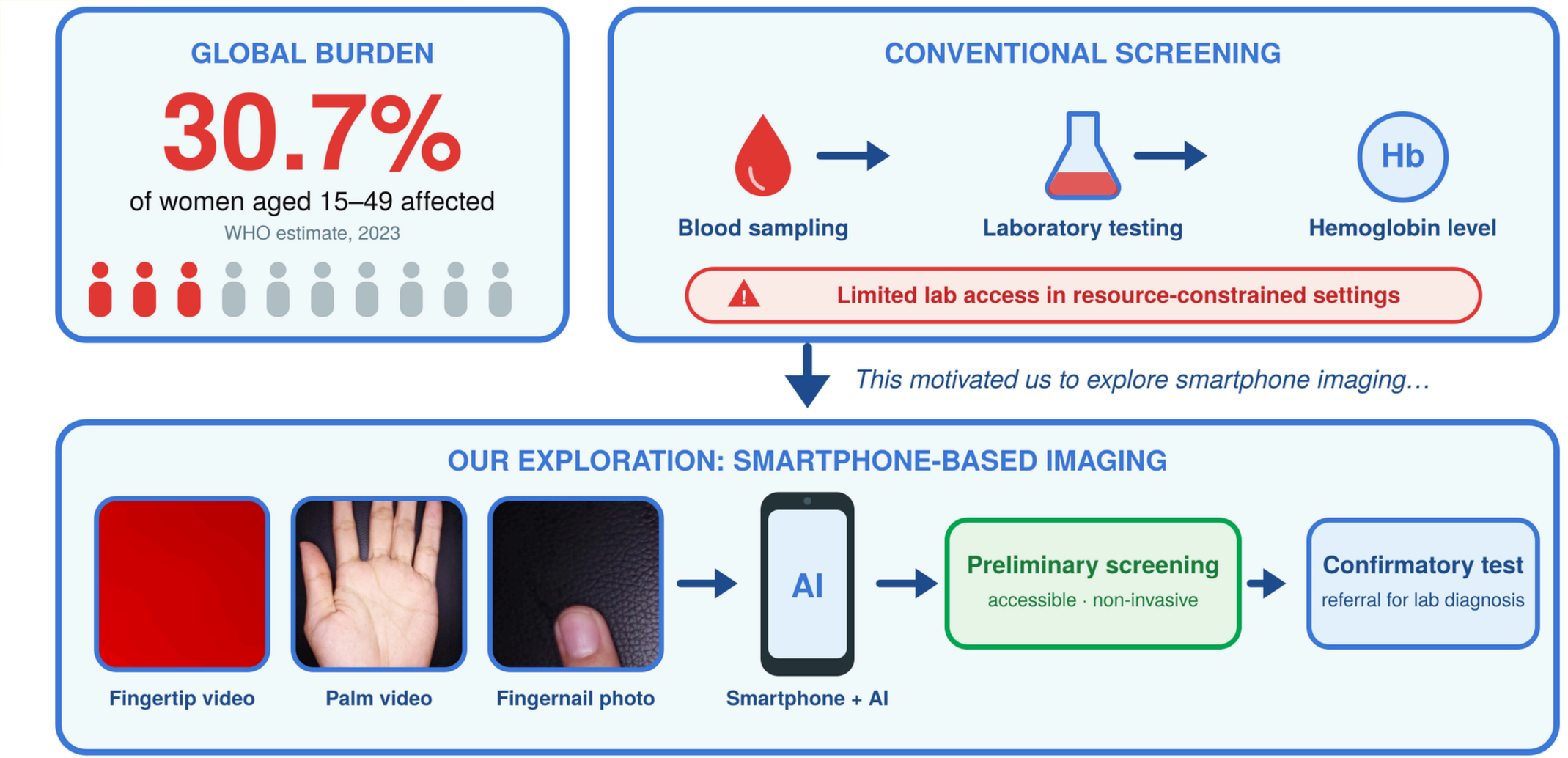
Possible anemia
→ get a lab test

Low risk
→ keep monitoring

What if a **simple smartphone camera** could help identify people who need **further testing**?

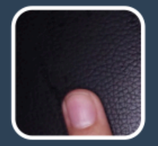
Easy · Non-invasive · Accessible anywhere

1.2 Screening Anemia with Technology: Background and Motivation



1.3 Anemia Detection from Body-Site Images: Existing Studies in the Literature

Research on Non-invasive Anemia Screening



Single-site images

- Das & Bhattacharjee, 2022
- Appiahene et al., 2023
- Sehar et al., 2025
- Navarro-Cabrera et al., 2025

Images + health records

- Ramzan et al., 2024
- Arul et al., 2026
- Yagnika et al., 2026

Multi-site photos

- Asare et al., 2023
- Nousir et al., 2024
- Mythili et al., 2024
- Lin et al., 2025

Fingertip video

- Wang et al., 2016
- Sabir et al., 2024
- Mannino et al., 2025

1.4 Limitations of Existing Research Studies

Few studies combine multiple sites, video and temporal modelling on a common benchmark.

Only one body site

Mostly nails or eyes alone

Appiahene et al., 2023 · Mythili et al., 2024
Kumar et al., 2025 · Krithika et al., 2024

Mostly still photos

Video rarely explored

Sabir et al., 2024

Limited fusion across sites

Joint use of sites is rare

Appiahene et al., 2023 · Mythili et al., 2024

No severity levels

Mostly anemic vs. not anemic

Appiahene et al., 2023 · Nousir et al., 2024
Mohammed et al., 2025

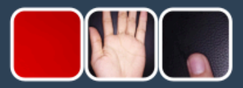
Small, rarely clinical data

~100–200 samples, few clinics

Mythili et al., 2024 · Sabir et al., 2024
Mohammed et al., 2025 · Das & Bhattacharjee, 2022

1.5 Research Objectives and Key Contributions

Research Objectives



To build one model using three body sites

Fingertip video · Palm video · Nail photo

To compare key design choices

Frames · Fusion · Clinical data

To compare with 11 published methods

Benchmarking

To evaluate with five-fold cross-validation

Robust evaluation

Key Contributions

Three sites in one model

Fingertip video + palm video + nail photo

Ablation study

Each design choice tested one at a time

11 methods re-run

Same dataset, fair comparison

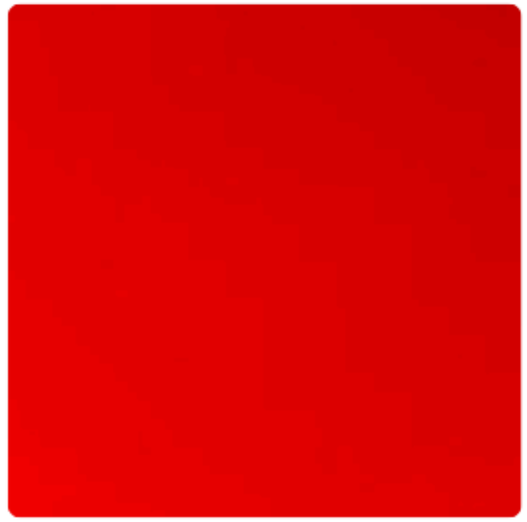
Unreliable result excluded

One experiment found and left out

2. Methodology

2.1 Multimodal Dataset

Dataset Name: Non-invasive Multimodal Dataset for the Detection of Iron Deficiency Anemia in Young Adults

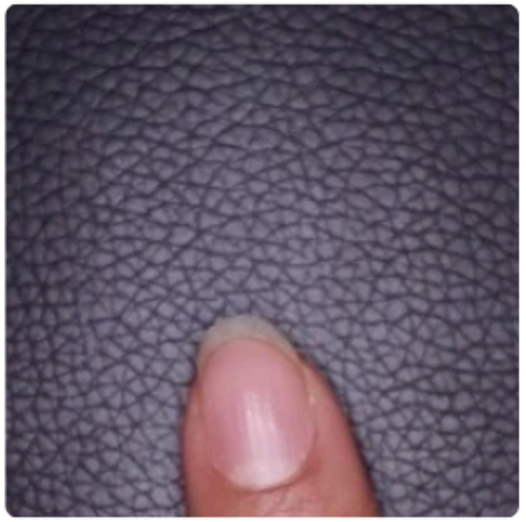


Fingertip video

30 s at 60 fps · right index finger

Palm video

Closed fist opened to a flat hand



Nail photograph

Light box · 2992 × 2992 px

909

young adults aged 18–25 · 540 women, 369 men

Participants per class

2.2 Preprocessing Framework

Preprocessing Information

Fingertip video **32** frames kept

Palm video **16** frames kept

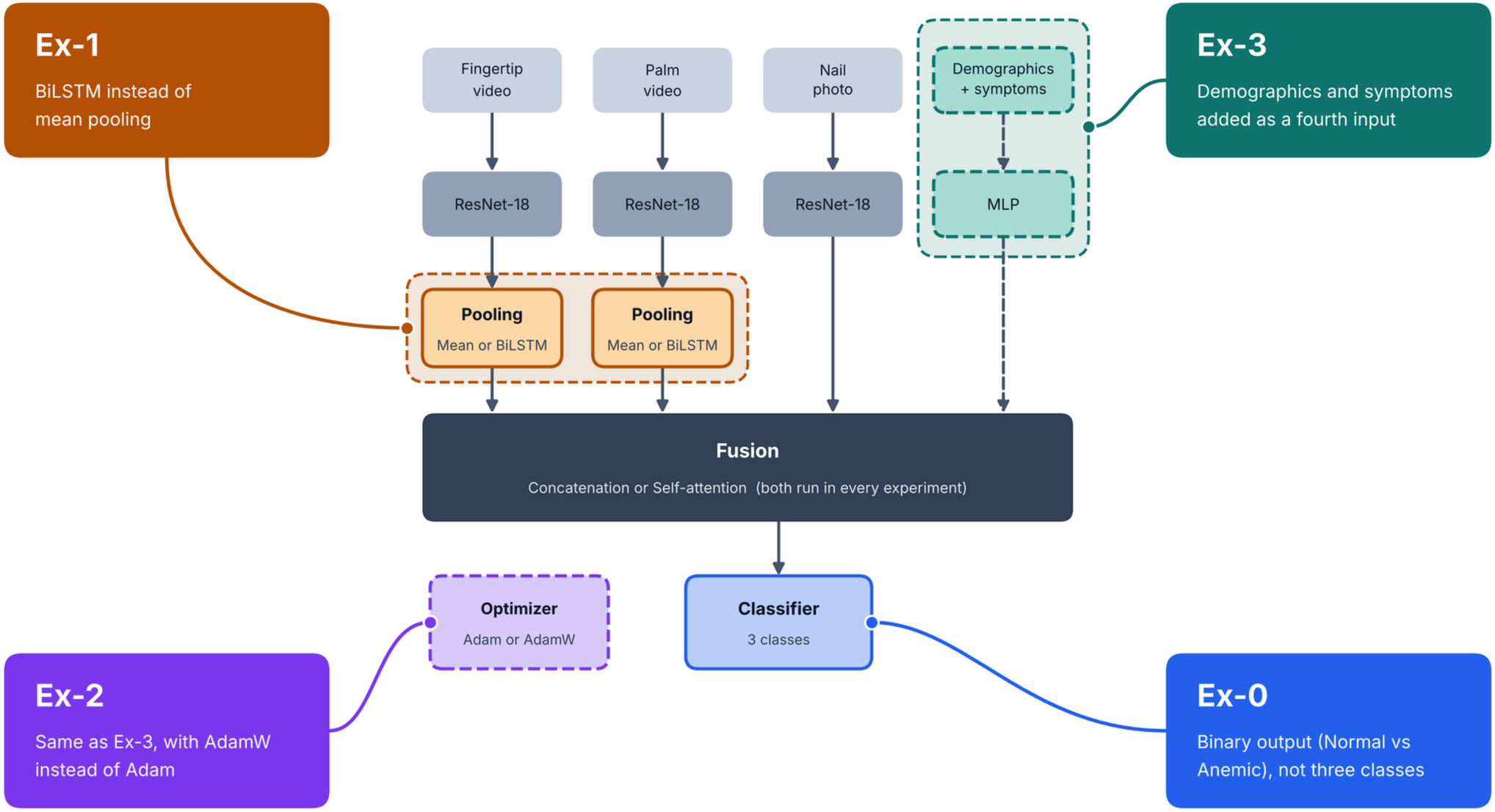
Nail photograph **1** crop

900 × 900 window

on the fingertip, shifted up 12%

2.3 Ablation Study Design and Experimental Configurations

(Ablation Configurations)



2.4 Our Model Architecture

model: ex-1: images only, with BiLSTM and self-attention.

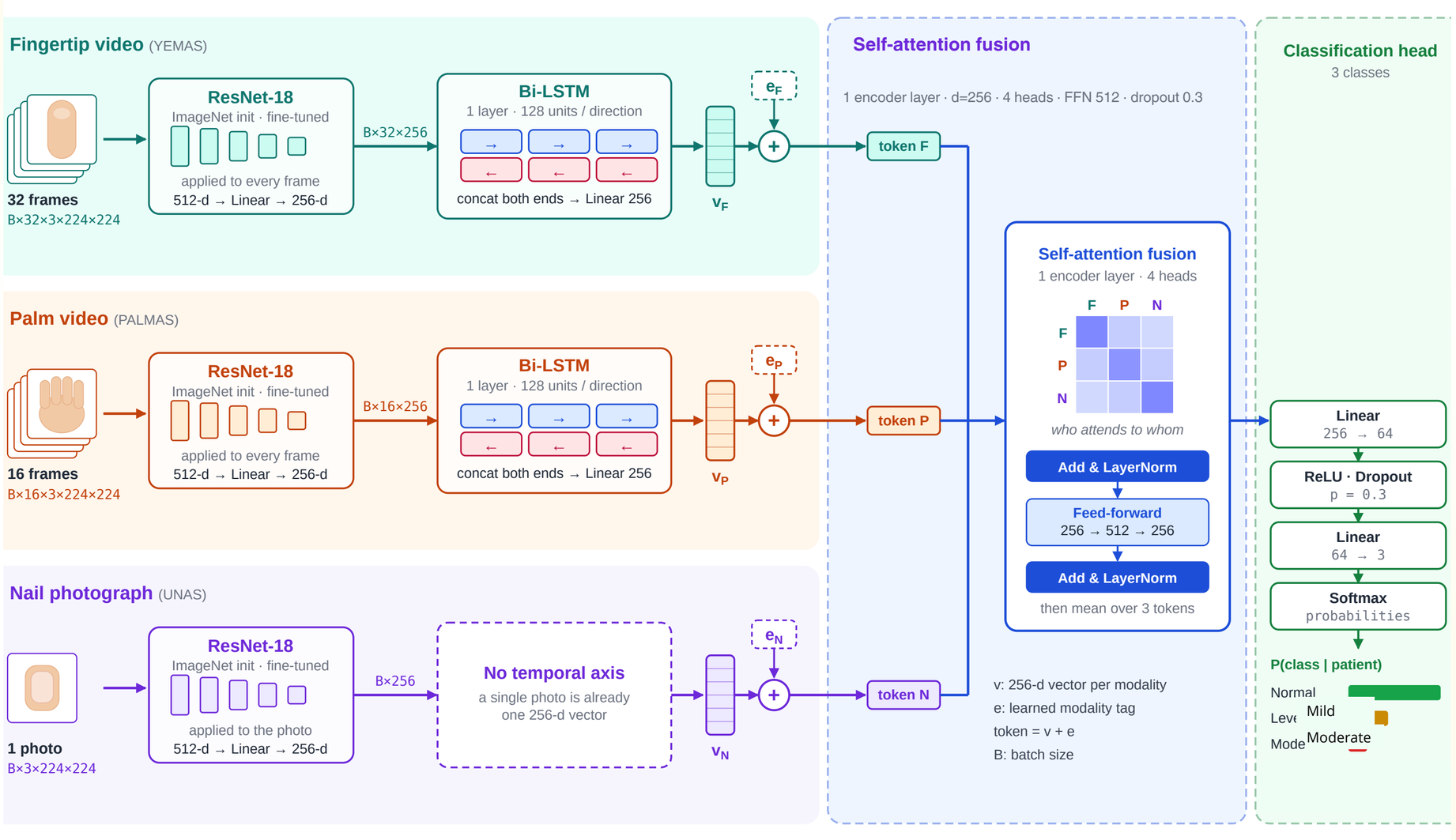


Figure: The model takes three inputs per patient: a fingertip video, a palm video and a nail photo. It turns each into a 256-number summary, lets the three summaries share information using self-attention, and then predicts the class (Normal, Leve or Moderada).

2.5 How Did Our Models Compare with Existing Approaches?

(Benchmarking)

1

11

published methods chosen

2

Re-built

each one re-implemented by us

3

Same data

our 909 participants, same 3 classes

4

Same score

macro-F1 for every model

3. Results and Discussion

3.1 Ablation Study Results and Performance Analysis

(Ablation results)

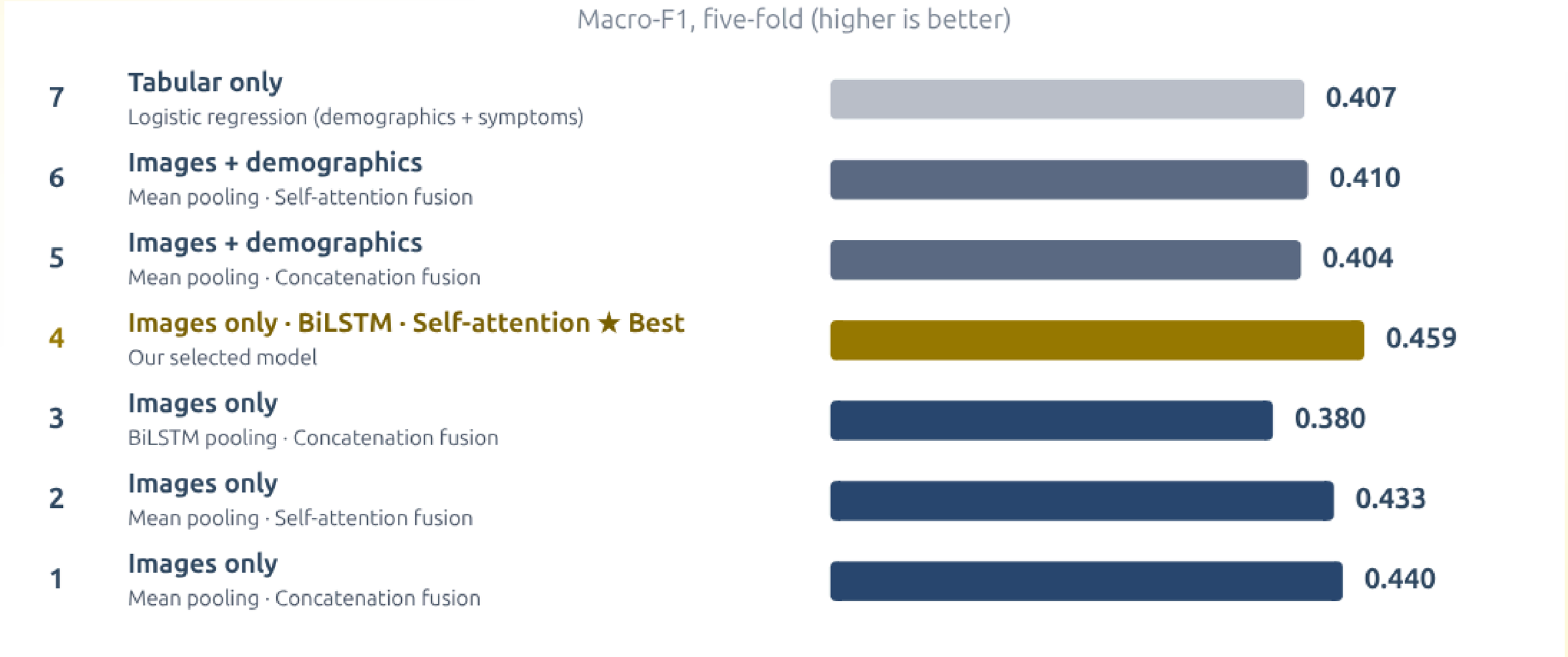


Figure : Ablation results (5-fold CV, mean ± SD). The highlighted configuration performs best.

3.2 Experimental Results and Performance Analysis

(Bench-marking Results)

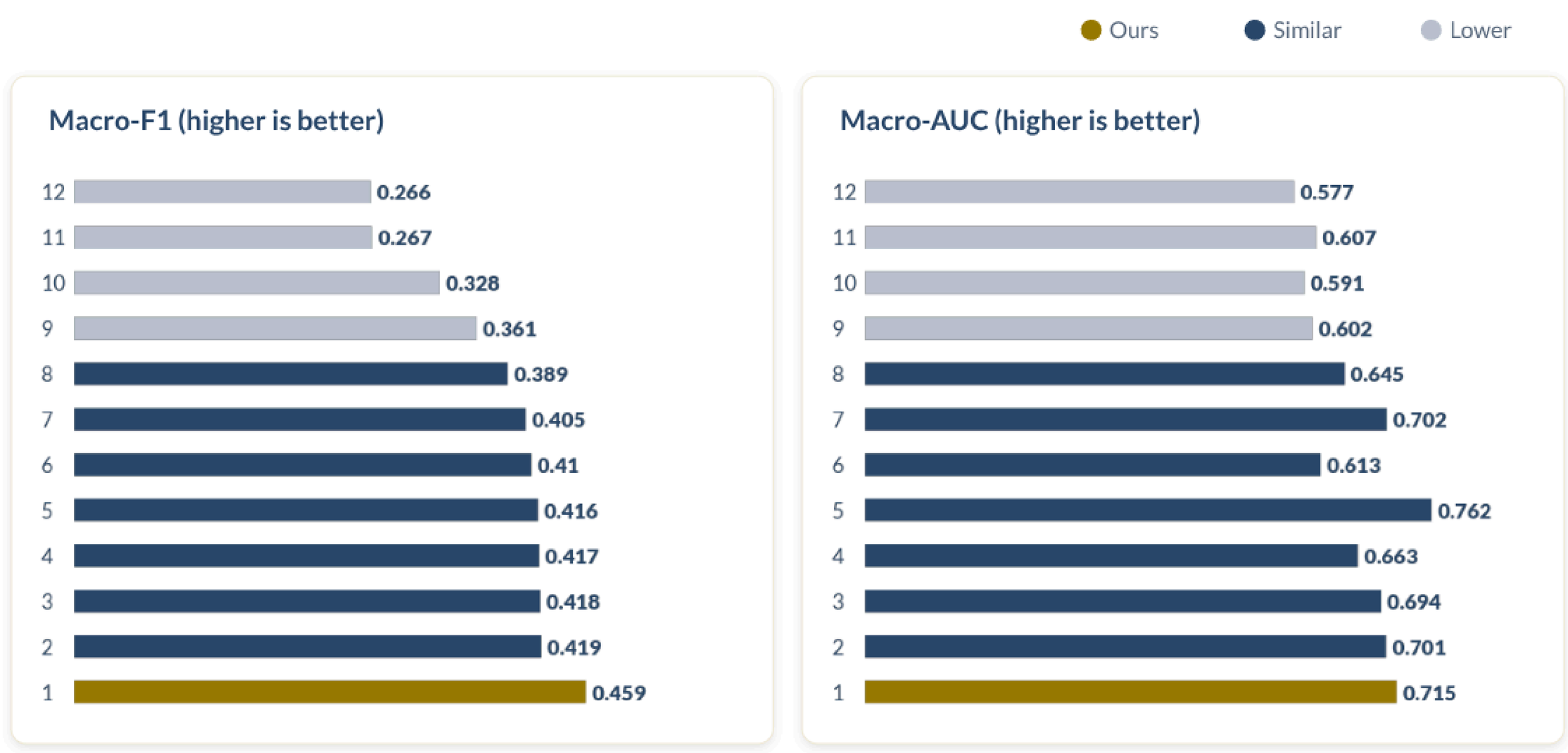


Figure : Macro-F1 and Macro-AUC Comparison of Benchmark Models

3.3 Interpreting the Experimental Findings

(Discussion)

What we found

Fingertip
73.87% accuracy (best)

Nail
67.27% accuracy

Palm
67.20% accuracy

Fusion models
macro-F1 0.380 to 0.459

≈

Tabular logistic regression
macro-F1 0.407

What the ablation showed: no gain

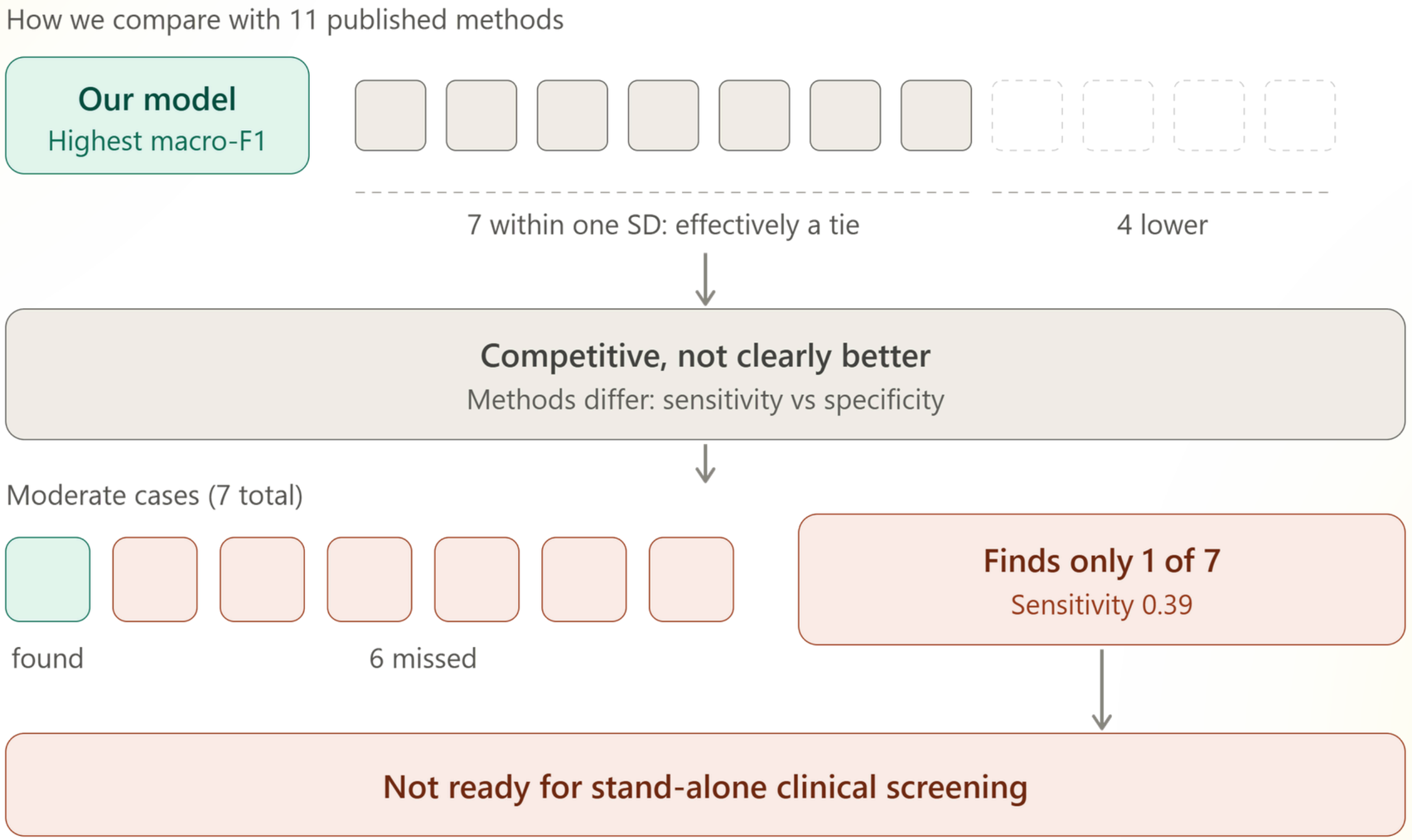
Self-attention
Not better than concat

BiLSTM pooling
No consistent effect

Demographics/symptoms
Did not help

3.3 Interpreting the Experimental Findings

(Discussion)



4. Conclusion

4.1 Conclusions and Research Implications

(Conclusions)

The signal is real
Fingertip, palm, nail all carry it

Fingertip video is strongest
73.87%; nail 67.27%, palm 67.20%

Fusion did not clearly help
F1 0.380 to 0.459 vs 0.407

No design choice mattered
Attention, BiLSTM, demographics

Competitive, not superior
Top of 11; 7 within one SD

Not clinically ready
Finds 1 of 7 Moderate (0.39)

Positive finding Neutral, no gain Caution

4.2 Limitations of This Study

Data	Test set	Comparison
One site only All 909 aged 18 to 25	**Only 7 Moderate cases** Less reliable recall	**Uneven testing** Five-fold vs single run across the 11 methods
No severe cases None in the dataset	**Found 1 of 7** Sensitivity 0.39	

4.3 Future Work

Collect more data

- **Wider coverage**
  Sites, devices, ages
- **More hard cases**
  Moderate and Severe

Improve the model

- **Moderate cases**
  Better detection

Validate fairly

- **Lab hemoglobin**
  Compare predictions
- **Repeated CV**
  Seeds and stat tests
- **Same rules**
  All 11, same tuning

References

Research paper: https://revistas.unsm.edu.pe/index.php/rcsi/article/view/955

Dataset Drive link: https://drive.google.com/drive/folders/1ztGpMMw0ni_VLqL353Ju34Kvw3WTmTye

- [1] E.-T. Lin, S.-C. Lu, A.-S. Liu, C.-H. Ko, C.-H. Huang, C.-L. Tsai, and L.-C. Fu, “Deep Learning-Based Model for Non-invasive Hemoglobin Estimation via Body Parts Images: A Retrospective Analysis and a Prospective Emergency Department Study,” Journal of Imaging Informatics in Medicine, vol. 38, no. 2, pp. 775–792, 2025, doi: 10.1007/s10278-024-01209-4.
- [2] P. Appiahene, J. W. Asare, E. T. Donkoh, G. Dimauro, and R. Maglietta, “Detection of iron deficiency anemia by medical images: A comparative study of machine learning algorithms,” BioData Mining, vol. 16, art. no. 2, 2023, doi: 10.1186/s13040-023-00319-z.
- [3] S. Mythili, R. A. Chelsia, and M. Thirishaa, “Anaemia Detection using Deep Learning by processing Conjunctiva, Fingernails, Palm and Tongue Images,” International Research Journal on Advanced Engineering Hub (IRJAEH), vol. 2, no. 9, pp. 2433–2439, 2024, doi: 10.47392/IRJAEH.2024.0333.
- [4] K. Kumar, A. Gupta, N. Gupta, M. Asim, and S. Srivastava, “Non-Invasive Hemoglobin Prediction: A Machine Learning Approach,” in Proceedings of the 2025 5th International Conference on Advancement in Electronics & Communication Engineering (AECE), 2025, pp. 620–625, doi: 10.1109/AECE67531.2025.11386530.
- [5] S. Krithika, A. K. R. Poreddy, T. Balakrishna, and P. Kokil, “A Multi-input Deep Neural Network Framework for Non-invasive Detection of Anemia using Finger Nail Images,” in Proceedings of the 2024 10th International Conference on Bio Signals, Images, and Instrumentation (ICBSII), 2024, pp. 1–6, doi: 10.1109/ICBSII61384.2024.10564094.
- [6] A. S. Nousir, A. M. Attia, M. Salama, A. Mostafa, M. Osama, H. A. Khalil, B. Mohamed, and D. Darwish, “Anemia Detection from Eyes, Palm and Fingernails with Machine Learning Models,” International Journal of Computer Science and Engineering Survey, vol. 15, no. 6, pp. 1–9, 2024, doi: 10.5121/ijcses.2024.15601.
- [13] R. Murugeswari, S. M. Kumar, T. H. Reddy, S. R. Reddy, and S. Eliyas, “Iron Deficiency Anemia Detection Using Deep CNN and Ensemble Learning,” in Proceedings of the 2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE), 2025, doi: 10.1109/ICCRTEE64519.2025.11052932.

References

- [7] H. Sabir, K. U. Khan, O. Ishaq, A. Alazeb, H. Aljuaid, A. Algarni, and J. Park, “Fingertip Video Dataset for Non-Invasive Diagnosis of Anemia Using ResNet-18 Classifier,” IEEE Access, vol. 12, pp. 68880–68892, 2024, doi: 10.1109/ACCESS.2024.3398353.
- [8] S. Das and M. Bhattacharjee, “Image-Based Sensing of Leukonychia for Early Diagnosis of Anemia Using a Smartphone Application,” IEEE Sensors Letters, vol. 6, no. 11, pp. 1–4, 2022, doi: 10.1109/LSENS.2022.3217010.
- [9] S. Kato, K. Chagi, Y. Takagi, M. Hidaka, S. Inoue, M. Sekiguchi, N. Adachi, K. Sato, H. Kawai, and M. Kato, “Machine/deep learning-assisted hemoglobin level prediction using palpebral conjunctival images,” British Journal of Haematology, vol. 205, no. 4, pp. 1590–1598, 2024, doi: 10.1111/bjh.19621.
- [10] J. R. Navarro-Cabrera, M. A. Valles-Coral, M. E. Farro-Roque, N. Reátegui-Lozano, and L. Arévalo-Fasanando, “Machine vision model using nail images for non-invasive detection of iron deficiency anemia in university students,” Frontiers in Big Data, vol. 8, art. no. 1557600, 2025, doi: 10.3389/fdata.2025.1557600.
- [11] K. K. Mohammed, N. Dahmani, R. Ahmed, A. Darwish, and A. E. Hassanien, “An Explainable AI and Optimized Multi-Branch Convolutional Neural Network Model for Eye Anemia Diagnosis,” IEEE Access, vol. 13, pp. 71840–71857, 2025, doi: 10.1109/ACCESS.2025.3560689.
- [12] A. A. Nababan, M. Turnip, A. Dharma, and Yennimar, “A Neural Network Optimization Approach for Anemia Prediction Using Image Pixels and Hemoglobin Data,” in Proceedings of the 2024 IEEE International Conference on Control & Automation, Electronics, Robotics, Internet of Things, and Artificial Intelligence (CERIA), 2024, pp. 1–6, doi: 10.1109/CERIA64726.2024.10915131.

Thank You

Questions and Answers